

RAHUL BAJAJ

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ENTRY LEVEL ARTIFICIAL INTELLIGENCE (AI) ENGINEER/DATA ANALYST

AI/ML specialist with expertise in building intelligent systems, LLM-driven automation, and scalable machine learning solutions. Skilled in leveraging large language models, prompt engineering, and cloud-based AI workflows to transform complex data into actionable, automated insights. Seeking an AI-focused role where I can design and deploy transformative solutions.

TECHNICAL SKILLS

Generative AI & LLMs: OpenAI GPT-4/4o, Claude Code, LLaMA 3, Mistral, RAG pipelines (LangChain, LlamaIndex)

Machine Learning: Scikit-learn, Random Forests, Gradient Boosting (XGBoost, LightGBM), Natural Language Processing (NLP)

General Skills: Python (Pandas, PyTorch), SQL, Tableau, Snowflake, Power BI, AWS, Databricks, Cursor AI

PROJECTS

Gemini AI & Streamlit: Next-Gen AI Blog Creator

- Developed an AI-powered blog generation platform using Gemini API, Python, and Streamlit, enabling automated transformation of text and image inputs into high-quality, contextually accurate content.
- Implemented secure API key management with dotenv, ensuring safe credential handling and smooth deployment across development and production environments.
- Designed and deployed an intuitive Streamlit interface for real-time content generation, delivering a seamless and efficient user experience powered by large language models (LLMs).

Vision AI: Real-Time Image Recognition with MobileNetV2

- Engineered an AI-powered image classification application leveraging MobileNetV2, OpenCV, and Pillow to deliver accurate, top-3 object recognition results directly from user-uploaded images.
- Built a robust image preprocessing workflow using OpenCV for resizing and normalization and Pillow for seamless image loading, ensuring consistent model input and reliable predictions.
- Developed an interactive Streamlit interface for real-time image analysis, improving accessibility and providing an intuitive experience for non-technical users.

EXPERIENCE

Datagurus, LLC.

Remote

Associate Analytics Engineer Intern

July 2025 - Present

- Collaborated on developing Beacon RAG v2, leveraging hybrid retrieval, re-ranking, and knowledge graph augmentation to improve enterprise AI search accuracy and relevance.
- Supported the implementation of advanced AI/ML techniques, including vector retrieval, semantic scoring, and multi-agent orchestration, enabling scalable, production-ready data solutions.
- Optimized and automated AI workflows, streamlining data ingestion, monitoring, and feedback loops, reducing query time by 25% and enhancing overall model performance.

Digital Infuzion Corporation

Rockville, Maryland

Clinical Data Analyst, National Institute of Aging (NIA)

October 2024 – April 2025

- Designed and implemented a standardized data dictionary covering over 100 disease and intervention terms based on NIH clinical taxonomy. Leveraged large language models (LLMs) to automate expert-defined clinical criteria mapping, reducing manual data processing time by ~35% and enhancing dataset uniformity for analysis.
- Redesigned, streamlined, and thoroughly reviewed a complex 5,000+ line Python/Pandas codebase, improving pipeline performance by 20%. Cut production workflow failure rates by over 15% and accelerated analytics delivery to support data-driven decision-making.
- Developed and facilitated 5 hands-on training sessions on prompt engineering strategies and LLM integration, directly impacting 20+ team members. Improved model query precision and reproducibility by ~20% and received an average workshop rating of 4.9/5 from participants.

Data Analyst, National Cancer Institute (NCI)

March 2024 - October 2024

- Redesigned NCI's cancer research dashboards using Plotly.js, transforming static charts into dynamic, interactive visualizations. Enhanced data clarity and user navigation, leading to a 25% improvement in researcher task efficiency and a 30% drop in follow-up

clarification requests.

- Developed advanced filtering and drill-down features within interactive dashboards, enabling real-time stratification of patient cohorts by biomarkers, demographics, and outcomes. Increased data exploration flexibility and accuracy by 20%, directly supporting more precise research hypotheses.
- Partnered with cross-functional teams of oncologists, data scientists, and IT specialists to identify and resolve critical visualization issues. Delivered scalable, research-ready dashboards that improved internal data reliability metrics by 20% and contributed to 2 ongoing clinical study platforms.

Junior Data Analyst & Curator, National Institutes of Mental Health (NIMH)

September 2023 - March 2024

- Transformed unstructured mental health data from 150+ participant surveys, experimental logs, and intake assessments into structured, analysis-ready formats. Streamlined data flow into centralized repositories, increasing accessibility and cross-study consistency by 40%.
- Constructed and validated complex datasets using advanced Excel functions and SQL queries, conducting integrity checks that reduced inconsistencies by 35% and supported accurate reporting for multiple longitudinal research projects.
- Liaised with 20+ healthcare providers, including hospitals, clinics, and behavioral health centers, to verify data compliance and resolve discrepancies. Improved data request fulfillment time by 20% through proactive communication and follow-up coordination.

EDUCATION

University of Maryland College Park

College Park, MD

Bachelor of Science, Information Science

August 2017 – May 2021

Minor in Technology Entrepreneurship

GPA: 3.60 / 4.00